Expected Goals (xG)

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1. Exploring xG Variations

Goal: Determine if different in-game situations have an effect on calculating xG.
Data Collected:
Tournaments: 2024 UEFA Euro Cup, 2024 Copa America, 2022 FIFA World Cup, 2021 UEFA Euro Cup, 2021 Copa America, 2018 FIFA World Cup
Variables: Country, Tournament, Round, Minute, xG, Score of game, Shot Type

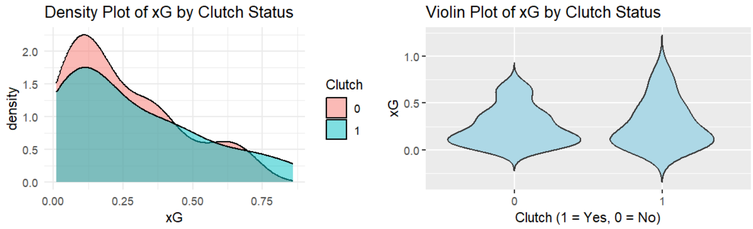
1.1 Clutch Goals

The first section of this project aims to explore the relationship between xG and late-game key moments, specifically clutch goals.

Definition. *A goal is considered clutch if it meets the following criteria:*
Any goal in extra time or in the final 10 minutes of regulation
To take the lead or tie the game

Determining Constraints: A *Chi-squared Test* was performed to assess whether the distribution of clutch goals across the game was significantly different from what would be expected by chance. The results (X^2 = 49.838, $p < 0.001$) revealed a strong association, indicating that clutch goals disproportionately occur in the final 10 minutes of gameplay.

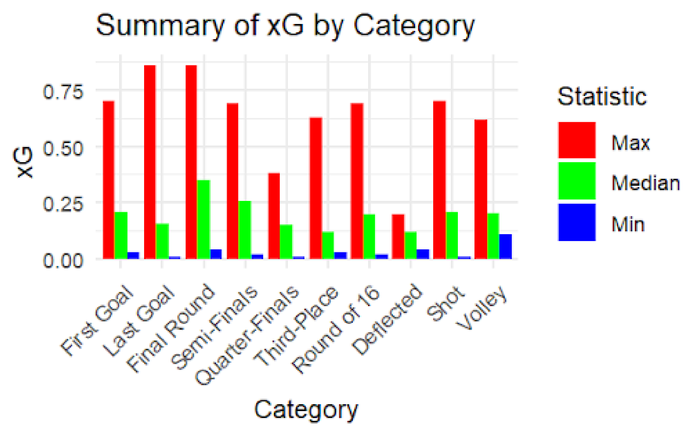
Results:



To confirm these observations, a *Welch two-sample t-test* was performed, which returned a *t-statistic* of -0.49495 and a *p-value* of 0.6244. These results indicate that the difference in xG between clutch and non-clutch goals is not statistically significant.

1.2 Other Potential Outliers

The next steps in the analysis involved identifying other potential outliers and anomalies in xG values by examining various factors and their minimum, maximum, and median xG values.



2. Modeling Shot Conversion Probabilities

Expected Goals:

Key soccer analytics metric that evaluates scoring opportunities by assigning a probability to each shot based on factors like shot distance, perceived goal length, and defender positioning.

2.1 Initial metrics for LR (Distance and Perceived Length)

Definition. *The perceived length is the visual angle of the goal as seen from the shooters perspective. We can extend this logic to 3D by including the height of the posts.*

Perceived Length (Representation in space)**.** *On the right is a drawing of how the perceived Length 3d is computed. It is defined as the product of the angles α and ϕ which are determined by using trigonometric identities. Adding the 3D angle phi was not beneficial for our model. The 2D version was kept for the version of our model.*

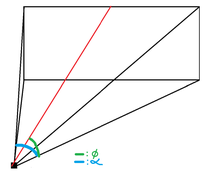


Figure 1: *Illustration of 3D Perceived Length*

Formula For 3D Perceived Length :

$$\text{PL3D} = \left(\frac{180}{\pi} \cdot \arccos\left(\frac{d^2_{\text{shot-to-left-post}} + d^2_{\text{shot-to-right-post}} - d^2_{\text{post-to-post}}}{2 \cdot d_{\text{shot-to-left-post}} \cdot d_{\text{shot-to-right-post}}}\right)\right) \cdot \left(\frac{180}{\pi} \cdot \arccos\left(\frac{d^2_{\text{top-to-shot}} + d^2_{\text{ground-mid-to-shot}} - d^2_{\text{top-to-mid}}}{2 \cdot d_{\text{top-to-shot}} \cdot d_{\text{ground-mid-to-shot}}}\right)\right)$$

2.2 Utilization of Freeze Frame

Freeze Frame: Coordinates of every player on the field at the time the shot was taken. Closest player in the triangle as defined in the 2d Perceived Length had the greatest chance of affecting the outcome of a shot and was considered in the model.

Freeze Frame (Representation of Freeze Frame)**.** *The distance between closest defender (points in red) and shot location is considered to have greatest impact on goal outcome. Goalkeeper position, point in green, is also considered in our model.*

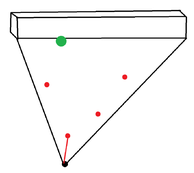


Figure 2: *Illustration of Freeze Frame*

2.3 Neural Network Training

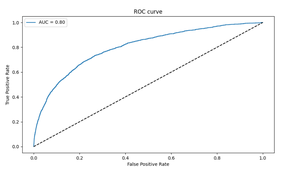


Figure 3: *ROC Score Graph*

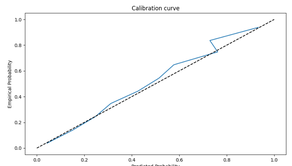


Figure 4: *Calibration Curve Graph*

3. Analysis of Shot Angle and xG Relationship

Research Goal: To develop an Expected Goals (xG) model analyzing shot location effectiveness in football, focusing on the relationship between shooting position, angle, and scoring probability.

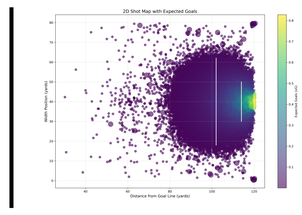
3.1 Methodology

Analyzed StatsBomb's open data format comprising 3,433 matches, yielding 87,111 total shots with 9,667 goals (11.1% conversion rate). After extracting shot event coordinates and outcomes, engineered key features including angle, distance, and field position metrics. Developed xG prediction model using logistic regression, with 80% data (n=69,689) for training and 20% (n=17,422) for validation, achieving 76% prediction accuracy.

3.2 xG Model Formula

Data Standardization

$$x_{std} = \frac{x - 103.571}{8.802}$$
$$y_{std} = \frac{y - 39.761}{9.895}$$
$$d_{std} = \frac{d - 19.203}{8.753}$$
$$\theta_{std} = \frac{\theta - 0.443}{0.276}$$



Formula: Probability Calculation

$$\log \text{odds} = -2.422 - 0.381x_{\text{std}} + 0.007y_{\text{std}} - 1.125d_{\text{std}} + 0.234\theta_{\text{std}}$$
$$\text{xG} = \frac{1}{1 + e^{-\log \text{odds}}}$$

Feature Impact: The model reveals shot distance as the strongest negative predictor (-1.125), while field position shows moderate negative influence (-0.381). Shot angle demonstrates positive impact (0.234), and width position exhibits minimal effect (0.007).

3.3 Key Statistical Findings

High-Probability Zones
Shots within the 6-yard box show highest xG (0.4-0.8)
Central penalty area shots average 0.2-0.4 xG
Penalty spot location records peak xG values of >0.7
Distance-Angle Relationship
Scoring probability decreases exponentially with distance
Wide angles significantly reduce xG regardless of distance
Optimal shooting positions cluster in central areas <12 yards from goal
Tactical Implications
High-value shooting positions strongly correlate with central penalty area
Wide positions beyond penalty area show consistently low xG (<0.1)
Data supports prioritizing central penetration in attacking tactics

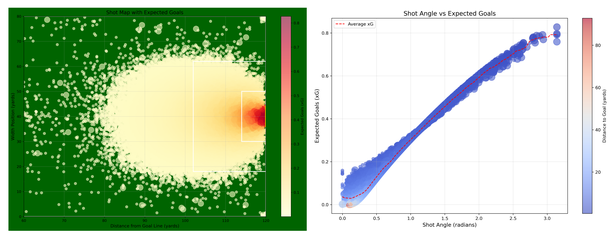


Figure 6: *Shot location heatmap displaying xG distribution across the pitch. Color gradient indicates scoring probability: red (high xG >0.4) to green (low xG <0.1). White lines denote penalty area and 6-yard box boundaries.*

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